C2 Amount of Substance – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Moles and Formulae **Papers 1 & 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Avogadro constant (N_A)				6.022×10^{23} mol^{-1}
2	Mole				Amount containing 6.022×10^{23} particles
3	Molar mass				Mass of 1 mol of a substance (g mol^{-1})
4	Equation linking moles, mass and M_r				$n = m \div M_r$
5	Equation linking moles, concentration and volume				$n = c \times V$ (V in dm^3)
6	Equation for number of particles				$N = n \times N_A$
7	Convert cm^3 to dm^3				÷ 1000
8	Units of concentration				mol dm^{-3}
9	Empirical formula				Simplest whole-number ratio of atoms of each element
10	Molecular formula				Actual number of atoms of each element in a molecule

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 2: Ideal Gas Equation **Papers 1 & 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Ideal gas equation				$pV = nRT$
2	Units of p				Pa
3	Units of V				m^3
4	Units of T				K
5	Value of R				$8.31\ J\ K^{-1}\ mol^{-1}$
6	Convert °C to K				+ 273
7	Convert cm^3 to m^3				$\times 10^{-6}$
8	Convert dm^3 to m^3				$\times 10^{-3}$
9	Convert kPa to Pa				$\times 10^{3}$
10	Molar gas volume at room temperature and pressure				$24\ dm^3\ mol^{-1}$

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 3: Yield, Atom Economy and Titrations **Papers 1 & 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Percentage yield equation				Actual yield ÷ theoretical yield × 100
2	Atom economy equation				M_r of desired product ÷ total M_r of reactants × 100
3	Benefit of high atom economy				Less waste
4	Limiting reactant				Reactant completely used up
5	Reactant in excess				Reactant left over at the end
6	Standard solution				Solution of known concentration
7	Concordant titres				Within 0.10 cm^3 of each other
8	Precision of burette readings				Nearest 0.05 cm^3
9	Ionic equation for neutralisation				$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$
10	Spectator ions				Ions that do not take part in the reaction

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C2 Amount of Substance – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 4: Water of Crystallisation and Uncertainty **Papers 1 & 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Water of crystallisation				Water molecules bonded within a crystal structure
2	Hydrated				Contains water of crystallisation
3	Anhydrous				Contains no water of crystallisation
4	Percentage by mass of an element				(A_r × number of atoms) ÷ M_r × 100
5	Convert mol dm^{-3} to g dm^{-3}				× M_r
6	Percentage uncertainty equation				Uncertainty ÷ measured value × 100
7	Uncertainty of one burette reading				±0.05 cm^3
8	Uncertainty of a titre				±0.10 cm^3
9	Number of readings in a titre				2
10	Glassware used to make up a standard solution				Volumetric flask

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.